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Heri Sutanta; Ni Putu Praja Chintya; Zhafirah Syarafina

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Issues and Challenges in Developing Geocoded Address in Indonesia

Heri Sutanta^{1, a)} Ni Putu Praja Chintya^{1, b)}, Zhafirah Syarafina^{1, c)}

¹*Department of Geodetic Engineering, Faculty of Engineering, Universitas Gadjah Mada, Jl. Grafika No. 2 Kampus UGM, Yogyakarta, Indonesia*

^{a)}Corresponding author: herisutanta@ugm.ac.id

^{b)}ni.putu.praja@mail.ugm.ac.id

^{c)}zhafirah.syarafina@mail.ugm.ac.id

Abstract. Location information, better known as geospatial information, is an indispensable element in city planning and management activities. It enables city planner and manager to develop optimum plan based on location and its spatial dependencies. Geospatial information supports the most detailed level of planning up to individual parcel and building. It supports management of city affairs, such as building permit and taxation, efficiently. Further, it also enables better service deliveries to those who request. In order to performing these activities well, geocoded address must first be developed. This paper investigates status, issues and challenges in developing geocoded address in Indonesia. The following aspects were examined: regulations, availability, base data to develop geocoded address, actors, and applications. A case study on developing geocoded address for a segment of Jalan Kaliurang in Yogyakarta for urban area and Baturetno Village for rural area were presented. This paper found that although geocoded address is a vital element for many development activities, it is still undeveloped.

INTRODUCTION

Development of houses, schools, offices, and other types buildings are a sign of a city's growth. Each of this object is built on a particular location, known as address. Address is a source of location information of a building which can be represented as a textual description or as a pair of coordinate. As the number of buildings increase, the number of address associated with those buildings will also increase which in turn make address representation become more complex. Building types also have some impact on address representation. Landed houses usually occupies a single land parcel which needs simple address representation. On the other hand, a unit in an apartment share land parcel with other unit which requires somewhat different method of representing address.

Address information is required in many applications, from sending postal package to managing city's utility services. With regard to growing interest in online shopping, geocoded address will be beneficial to package delivery business. In addition, travel industry which relies on address to pick up passenger will also be facilitated. Traditionally, an address is represented using an agreed form of description, typically using street number, municipality name, and followed by postcode. This method has been used since long time and is the most popular way of describing location information. However, in the framework of geospatial information system (GIS), an address of a particular location needs to be enhanced by adding geolocation. It is a process known as geocoding, a process to give a pair of coordinates in an address [1]. It requires address data and reference data. Although geocoded address is essential in delivering many types of city's service, geocoded address has not been developed in Indonesia, nationally or locally.

Geocoding has been applied in many fields that involves analysis, visualization, planning and exploration of data¹. Application that involve spatial analysis needs geocoding result which has a high accuracy [2], for example, analysis of crime [3,4], research on cancer [5,6], social inequalities [7], and government [8].

Indonesia has yet to develop national standards on address. Address descriptions currently in use are varies. Significant differences can be seen in the structure of address in urban and rural areas. Address in rural areas usually uses the following addresses components: Rukun Tetangga (RT) number, name of the kampong and village name, and sub-district name. Urban areas typically use street name and house numbers as the main component of an address. Differences in the form of address create difficulties to transform textual address into spatially referenced database.

GEOCODING

Geocoding Methods

There are three commonly used methods in geocoding: street network, point, and parcels [9]. These methods uses differents data model. Each data model has a different geometry types and format in the representing address. Geocoding using line based reference data is a traditional method, but is the most popular method especially for urban area. Street network data includes a street address which associated with buildings such as homes, offices, and other utilizations. This concept was first introduced in TIGER, a data model developed by the US Census Bureau [10]. This data model relies on a range of street numbers associated with the centre line of the street segment [11].

The point-based method, also known as point-address, requires point features data reference. The point reference data be obtained through national mapping agency or national land agency. It can be constructed from parcels or buildings. The area-based method, known as parcel geocoding, is based on putting address on the area of the polygon. The result of an area-based geocoding is a point feature based on interpolation. The output reflects the centroid of the polygon or in the area of a polygon which mean that these results are influenced by the shape of a polygon. The larger and more irregular shape of a polygon, the more inaccurate the result of interpolation.

Geocoding Platforms

There are several tools that can be used to perform geocoding, whether it is proprietary software, open source, as well as desktop based and web-based geocoder. Examples of proprietary software are Centrus US Street Point Database, Geolytics GeocodeDVD, ESRI Address Locator. Examples of a web-based software are Geocoder.us, Google Earth, Google Maps API (currently is not active any more) and Yahoo API and example of open source software is USC Geocoding Platform [12].

ESRI's ArcGIS family that can be used for geocoding are ArcMap and ArcCatalog. ArcCatalog serves to collate reference data, create the address locator, geocoding addresses and publish address locator. ArcMap can be used to display the results geocoding, conducting geocoding interactively, and review of data results in an interactive geocoding, provides toolboxes, aligning with the address locator with reference data, and analyze the results of geocoding.

Geocoding Algorithm

Geocoding involves two basic processes: feature matching and feature interpolation¹³. Feature matching is a process to identify a geographic feature in the reference data set corresponding to the input data that will be used to produce the final geocoding result. Feature interpolation is a process of obtaining or retrieving the data output coming from geographic feature data obtained during features matching process. Interpolation is the process of generating geographical features from the reference feature, for example, produces a point along the street or centroid plots of parcels or buildings.

Two types of algorithms in features matching are interactive and non-interactive matching algorithms [13]. Interactive matching algorithm is an interactive feature matching process, that is when the process of feature matching failed, users can correct input data. Non-interactive matching process is done automatically without involvement of an operator.

The quality of geocoding results indicated by completeness, positional accuracy and repeatability [14]. Completeness is determined by the level of overall compatibility, which refers to the percentage of data that can reliably gocoded. It is is often referred as a match rate [15]. Accuracy refers to how close the position of each geocoding results with the actual location of that address. Repetition shows how sensitive the geocoding results to

variations in the input address, reference data, matching algorithm software, the ability and interpretation analysts. Quality of the reference data and the input data affect the quality of geocoding [14].

ISSUES AND CHALLENGES

The need for geospatial data is ever increasing. Geospatial data are required in many application areas, from planning city's development, analysis of service areas, to develop evacuation route in the event of a disaster. Technological developments also take part in increasing the need of spatial data. Non-spatial data need to be integrated with spatial data to form new spatially related information. People will find it is easier to understand textual data than coordinate data, and map is better in visualize information than textual message. Address is textual information that shows a geographic location, however if an address is represented as a pair of coordinate, it will be difficult to find out where on earth the location is. On the other hand, geospatial information software needs coordinates to depict location in the database. Geocoding process is necessary to transform textual data into, and link with, spatial data. We identified several issues related to geocoding: (1) delivery or routing services, (2) reference and address data provider and (3) government programs.

Addressing in rural areas usually use the addresses components such as the name of the village, kampong names, numbers of Rukun Tetangga (RT) and the number of Rukun Warga (RW). A postal package is not delivered directly to the recipient, but to the village office. The village staff will then deliver the package to the kampong head or the recipient can get it in the village office. This process is time consuming and important package may delivered late to the recipient. Geocoded address is needed to accelerate this process, which facilitates the delivery service officer deliver the packages directly to the recipient in time and to the right location. Geocoding also enables companies to perform route sequencing technology, which calculates the most efficient way to deliver or visit multiple locations. Routing services is also essential in business to ascertain proximity to potential customers, distance to suppliers and competitors, service areas and delivery routes.

Address data and reference data are main components to perform geocoding. Availability of both data in Indonesia was inadequate. Indonesia does not have a specific agency responsible for making regulations and managing address data. This causes variations and redundancy in address data. There are many agencies that provide reference data, however with different standards and methods.

Several government programs require address information to ensure the program can be run effectively and efficiently. The data required are personal data and geocoded address. Examples of programs that require geocoded data are census, elections, and the distribution of direct aid. In census program geocoded address data are needed to determine the distribution of the population. In the distribution of direct aid, geocoded address data were used to facilitate the distribution government aid effectively and efficiently.

Geocoding result depends on input data from various sources which many factors need to be considered. Implementation of geocoding in Indonesia faces several challenges, such as (1) availability of reference data, (2) incomplete reference data, (3) data provider, (4) suitable address locator, and (5) the up-to-date-ness of the data.

The reference and address data are not complete. Although there are many agencies that conduct geospatial data acquisition, there is deficiency in the availability of parcel level map necessary for geocoding. Parcel level map is one key element in developing geocoded address, however four situations found as impediment in the utilization of it. First, the available parcel maps covers less than 30% land parcel in Indonesia. Second, many land parcels have not been mapped (although they might have been surveyed), leaving holes/empty spaces between the mapped land parcels. Third, the currentness of the available land parcel cannot be guaranteed since the updating process were infrequent and limited metadata available on it. Four, accessing land parcel map is not easy, especially in navigating the bureaucratic processess.

In case the required data exist, which is rarely found, different agencies may have data with different quality. In effect, combining data with different quality may result in erroneous address. A national agency acting as custodian in this topic need to be defined, as currently no government agency is responsible for developing national standard for addressing as well as acting as the custodian for address data.

Address locator is a tool to transform address data to spatial data. Each address locator have specific rule depending on the reference and address data. The rule contains the format of the address, type of reference data and feature matching algorithms. Common address locator developed overseas may not suitable, and directly applicable, to be used in Indonesia. An address locator that suitable with Indonesian specificity, address conditions and reference data, needs to be developed.

The reference and address data available is commonly not up to date. Updating process is costly and time consuming, and therefore infrequently conducted. The situation leads to a mismatch between address and reference data. This mismatch causes poor quality geocoding results and might not reflect the latest field condition.

CASE STUDY

To investigate the possibility to develop geocoded address in urban and rural setting in Indonesia, two case studies were conducted. For urban area, a street segment of Jl. Kaliurang was chosen ¹⁶, while for rural area a Baturetno village in Kabupaten Bantul was selected ¹⁷. Jl. Kaliurang is a typical city street with buildings built next to each other. The segment stretches from Selokan Mataram to North Ring Road. Many buildings were used for multiple purposes, some were identified as have been sub-divided from single land parcel. Only few empty parcel found, as they are being in the development phases or waiting to be developed. Two kampongs in Baturetno were chosen, Wiyoro Kidul and Bumen Kulon. Baturetno village is located in Kecamatan Banguntapan, Kabupaten Bantul, see The kampongs located in urban fringe but with village characteristics still dominated the area.

Data and Software

The data used in this investigation were reference and address data. For Jl. Kaliurang, the reference data used were area-based reference data (building features) from National Land Agency (BPN) Sleman. The data was created in 2005. Additional data were obtained from OpenStreetMap (OSM). Address data were obtained from field survey as well as website and social media, which were conducted in April 2015. For Kampong Wiyoro Kidul and Bumen Kulon, three reference data were obtained, Block Map developed by the (former) Directorate General of Tax, Land Parcel Map by BPN and OpenStreetMap data. The reference data from OpenStreetMap was chosen because the reference data from BPN were incomplete and the reference data from the Directorate General of Tax did not match with current situation in the field. The following software were used in geocoding: JOSM and ArcGIS.

Existing Address Characteristics

An address consists of combination of the components of the address. The typical components of an address used in urban area are street name, alley's name, house number, kampong name, village name, sub-district name, district name, province name and zip code. Address conditions found at urban areas have characteristics as shown in Table 1. The numbers of buildings used in the study were 186 units. Figure 1 shows address conditions in Jalan Kaliurang. In Fig. 1a., four buildings share the same address (Apotek K24, Photo Talk, Aisya, Acaciana), while Fig. 1b. depicts incomplete address as several builings only show the km of building (e.g. Luxury, Jalan Kaliurang km 4.5). In rural area, the following address components were used: house number (if any), Rukun Tetangga (RT), village name, sub-district name, district name and postcodes. Five RTs were selected as the study area, RT 05, RT 06, RT 07, RT 08 (Kampong Wiyoro) and RT 09 (Kampong Bumen Kulon). An address can be defined as complete if they consist of house number. In the study area, majority of the address were incomplete. Out of 311 buildings, 184 have incomplete address. Table 2 shows the situation in Baturetno.

TABLE 1. Addresses conditions in Jl. Kaliurang.

Condition	Number	Example
Complete addresses	90	Jalan Kaliurang Km 5.6 No 25
Incomplete addresses	96	Jalan Kaliurang Km 5.3
Single address used for more than one building	8	Jalan Kaliurang Km 5 No 94 (Acaciana and Photo Talk)
Addresses with sub number	11	Jalan Kaliurang Km 5.6 No 17A
Addresses of the building located at the intersection of two streets	2	Jalan Kaliurang Km 5, Karangwuni, Blok A No 1

TABLE 2. Addresses conditions in Baturetno.

Address condition	RT 05	RT 06	RT 07	RT 08	RT 09	Total	Example
Complete addresses	39	17	0	71	0	127	Wiyoro Kidul RT 05 No 174
Incomplete addresses	7	35	34	17	91	184	Wiyoro Kidul RT 06
Total	46	52	34	88	91	311	



FIGURE 1. Single address for more than one building (left), incomplete addresses (right)

Geocoding Process

The geocoding process followed the standard process in ArcGIS, as illustrated in Fig. 2. Geocoding framework consists of geocoding user interface, address locator and rule base for matching. Geocoding user interface provides an interface for the user to perform geocoding. ArcGIS provides two main application that serves as an interface that is ArcMap and ArcCatalog. Address locator a combination of reference data and selected type of address locator. Address locator was made in ArcGIS user interface. Rule base is a collection of files that were used to translate the address data into spatial data.

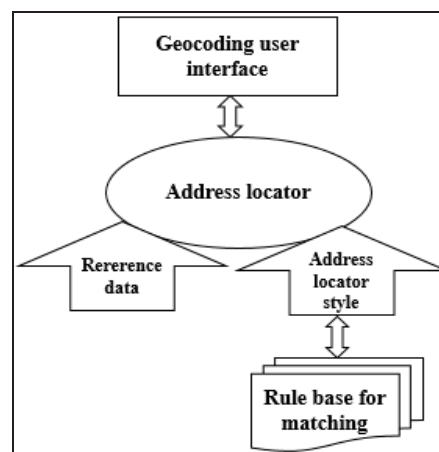


FIGURE 2. Geocoding framework in ArcGIS ¹⁸

The geocoding process begins with the preparation of address data and reference data. The reference data were building footprints, which in case study in the urban area was obtained from BPN and in the rural area from OSM. Address data were obtained through field surveys and internet browsing. The second stage was decomposed of address data to determine the components of each address. The components of each address were used as an attribute of the reference data. A possibility of errors in writing address was checked manually. A typical errors include abbreviating address and similar address for two different locations.

The next step was creation of an address locator. An address locator contains reference data that was equipped with address data. It was followed by geocoding addresses itself. When conducting geocoding process there were feature matching process and features interpolation process. Geocoding process returned points with specific coordinates inside a particular features (e.g. building) to be assigned geocoded address.

Geocoding Result

The geocoding method used in this research was Area-based Geocoding. Feature matching algorithm used in the area-based geocoding method is probabilistic algorithms. Output features were not only classified as matching or unmatching, but was also as tied result. Tied result generated because of an error in writing address or single address data for more than one features. Geocoding matching result for area based reference data using two types of address locators, the General Single Field and US Address Single House, are presented in Table 3. General Single Field is an address locator that allows the use of a key column for a match between the reference data and address data. Each column is identified by name or code. Key column used in this study is the name of the building. US Address Single House is an address locator that requires reference data in the form of a polygon or point features.

TABLE 3 Geocoding results		
Location	Address Locator	Matching Result
Urban Area	General Single Field	170
	US Address Single House	65
Rural Area	General Single Field	105
	US Address Single House	92

Figure 3 and 4 show maps of the result of geocoding in urban and rural area using two types of address locators, respectively. Figure 3a. is the result of geocoding use General Single Field address locator and Fig. 3b. is the result of geocoding use US Address Single House. The matching rate of Fig. 3a. is bigger than Fig. 3b. because most of building in urban area have a unique name and do not have a complete address.

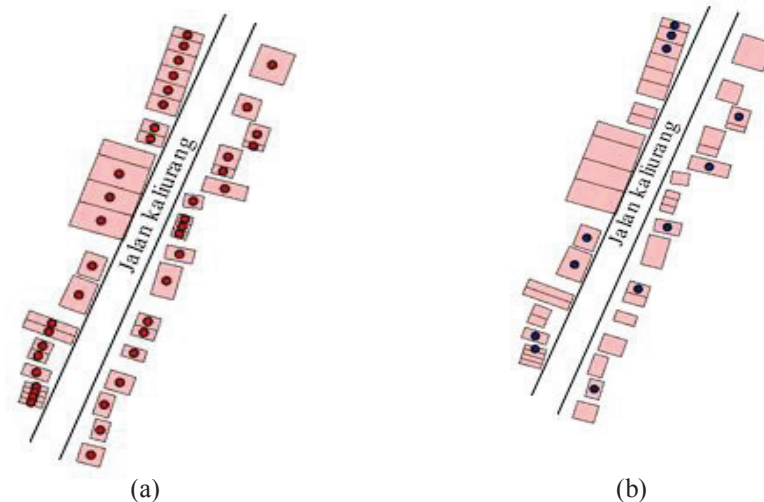


FIGURE 3 Geocoding result in urban area using General Single Field (a), and US Address Single House method (b).



FIGURE 4. Geocoding result in rural area using General Single Field (a), US Address Single House method (b).

The highest match rate was the result of geocoding use General Single Field address locator for urban and rural areas. This was caused by situation that many buildings do not have a complete addresses. Geocoding would be successful if there is good quality of address data. Address data in this study has many deficiencies, such as incomplete address, house number was not sequential, addresses with different format, and kilometer reference was not sequential.

The reference data used to perform geocoding in this study was building data. Ideally, each building has an address that is different from the other buildings. Building reference data for urban areas obtained were digitized data from aerial photographs in 1997. Consequently, there were many differences with the current conditions. The differences encountered create problem in the implementation of the. The building reference data for rural areas were obtained from digitization on OpenStreetMap because the data from the government in these areas were not suitable to be used as reference data.

US Address Single House and General Single Field were not suitable for address that did not have a house number, as it create a non unique address. The result of geocoding process for incomplete address was a tied address. Tied addresses were caused by addresses with same number that located in different *Rukun Tetangga*. This address locator did not support component address like *Rukun Tetangga*. General Single Field result was similar to US Address Single House because in rural area most of building were houses. Houses in rural area did not have address structure similar to building in urban area.

CONCLUSIONS

Geocoded address is essential in facilitating many types of city service delivery. It will also contribute to the creation of supportive environment for online market which currently is increasing substantially. Development of geocoded address in Indonesia has to address several issues and faces many challenges. The absence of a uniform address model and the availability of land parcel map impeding the development of geocoded address. The biggest challenge for Indonesia to develop geocoded address is to provide address and reference data in a standardized format. Address format currently in use in many cities is varied and show some dissimilarity with that of overseas. Common address locator in the market cannot produce high quality geocoded address. Hence, suitable address locator has to be created.

Based on the case study, address locator method which can be used in Indonesia are General Single Field and US Address Single House. The best result was found in General Single House address locator. Characteristic of the reference data highly affect the geocoding results. Reference data owned by the government need to be completed and updated to facilitate the development of geocoded address.

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